

Md. Meheraj Hossain

Dhaka, Bangladesh | +880 1684251757 | meherajhossainmahir@gmail.com
linkedin.com/in/md-meheraj-hossain | github.com/mhstyles7 | meherajportfolio.vercel.app

Professional Summary

Computer Science and Engineering graduate from BRAC University with a published first-author research paper in ICT Express (Elsevier, Q1, IF: 4.2) on AI-driven zero-day intrusion detection using Siamese Networks and Reinforcement Learning. I build full-stack web applications using the MERN stack and work across machine learning, cybersecurity, and IoT systems. My projects are live and deployed — not just academic exercises. Currently open to opportunities in software engineering, AI/ML, corporate IT, and research roles in Bangladesh and beyond.

Education

Bachelor of Science in Computer Science & Engineering	2021–2026
BRAC University, Bangladesh	CGPA: 3.61/4.00
Higher Secondary Certificate (HSC)	2018–2020
Mirpur Cantonment Public School & College, Bangladesh	GPA: 5.00/5.00
Secondary School Certificate (SSC)	2016–2018
Greenview High School and College, Bangladesh	GPA: 5.00/5.00

Technical Skills

Systems & Security: Linux, Wireshark, Network Protocol Analysis, MySQL, Git

Web & Application: MERN Stack, Next.js, REST APIs, JWT Authentication, MVC Architecture

Languages: Python, JavaScript, TypeScript, SQL, C

Tools & Deployment: Postman, Vercel, Render, TensorFlow, Scikit-learn

Projects

CollabBD – Bangladesh’s Talent Network *Apr 2026 – Present*

Solo Project · Live: collab-bd-server.vercel.app · GitHub: github.com/mhstyles7/CollabBD

- Engineered a full-stack, real-time talent marketplace (Next.js, TypeScript, Node.js, Express, MongoDB) featuring Socket.IO-powered messaging, live community rooms, and geospatial talent discovery, secured with rate limiting, NoSQL injection protection, Helmet headers, JWT authentication, and an admin-moderated ID verification and proposal-bidding system.

Trippy 2.0 – Travel Social Media & Tour Management System *Apr 2026 – May 2026*

Solo Project · Live: trippy-2-0.vercel.app · GitHub: github.com/mhstyles7/Trippy-2-0

- Rebuilt a full-stack MERN application end-to-end, integrating OCR-based automated ID verification, secure JWT role-based access, and Postman-validated REST APIs built on MVC principles for scalable, maintainable backend operations.

PothChola – Localized Smart Travel and Cultural Discovery Platform *Dec 2025 – Feb 2026*

Live: poth-chola.vercel.app · GitHub: github.com/mhstyles7/PothChola

- Led React frontend development and MongoDB schema design for a team-built MERN app supporting complex group itineraries and social features, integrating Gemini AI and Google OAuth, and managing full-stack cloud deployment across Vercel and Render.

Air Quality Monitoring System (ESP32-Based IoT) *Dec 2025 – Jan 2026*

Details: meherajportfolio.vercel.app/#project-04

- Built a multi-sensor (MQ-series gas, DHT11) air quality monitoring system on ESP32 with automated MOSFET-controlled exhaust fan actuation, SD card data logging, and a Wi-Fi-streamed live monitoring dashboard — hands-on experience with embedded systems and hardware-software integration.

Research

A Cross-Dataset based Zero-Day Intrusion Detection System by Integrating Siamese Network and Reinforcement Learning

First Author · Accepted in *ICT Express* (Elsevier, CiteScore: 10.8, IF: 4.2) · doi.org/10.1016/j.ict.2026.05.001

- Designed and led an AI-based network intrusion detection system targeting zero-day cyber threats across heterogeneous IoT environments — directly applicable to enterprise network security and threat monitoring.
- Applied network traffic analysis and adaptive machine learning models (PPO/DQN/SAC) to detect anomalous behavior, demonstrating deep understanding of network security principles.
- Presented this research alongside a working proof-of-concept prototype at the inaugural North South University Cyber-security Center (NSU CSC) Showcase, held in the presence of **Mr. Rehan Asif Asad, Advisor to the Honorable Prime Minister (ICT and Science & Technology)**. The solution received positive feedback and generated interest from government representatives, researchers, and cybersecurity professionals.

Details: meherajportfolio.vercel.app/#nsu-showcase

Extracurricular & Additional Experience

Graphics Design (Beginner Level) *Sep 2022 – Dec 2022*

Extracurricular Activity, BRAC University

Social Media Manager *Feb 2024 – Jul 2024*

Bangladesh Freight Forwarders Association (BAFFA)

- Managed digital communications for a senior executive holding five concurrent leadership roles across international freight, logistics, and trade associations; maintained a platform with over 68,000 followers, aligning content with institutional objectives and managing responsibilities independently.

Competition Participant & Awardee *2018*

IAM Digital Programme, conducted by Dr. Mohammad Kaykobad, Professor of CSE, BUET · Mirpur Cantonment Public School & College